



Faculty of Engineering

School of Electrical Engineering and Telecommunications

ELEC 1111 – Topic 3

Circuit Theorems

Dr. Inmaculada (Inma) Tomeo-Reyes

Senior Lecturer

School of Electrical Engineering and Telecommunications, UNSW

Topic 3 Content

This lecture covers:

- Superposition
- Source Transformation
- Thevenin's and Norton's theorems
- Maximum Power Transfer

**Corresponds to Chapter 4
of your textbook**

Addressed learning outcomes:

- *LO1*: Systematically analyse DC and AC electric circuits by deriving and solving equations using circuit laws and theorems.
- *LO6*: Evaluate relevant information to design simple engineering systems that use electrical circuits.

Superposition

If there are **two** or **more independent** sources in a circuit, there are different ways to solve for the circuit parameters:

- Nodal or mesh analyses.
- **Superposition.**
 - It is based on the **linear property** of circuits.
 - Only **one independent** source is considered at a time.
 - The **rest** of the independent sources are **set to zero** (turned off).
 - **Dependent** sources are **left intact** since they are controlled by circuit variables.

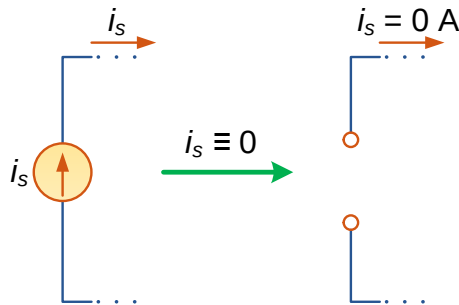
The **superposition** principle states that the voltage across (or current through) an element in a **linear circuit** is the **algebraic sum** of the voltages across (or currents through) that element due to each **independent source** acting **alone**.

Superposition

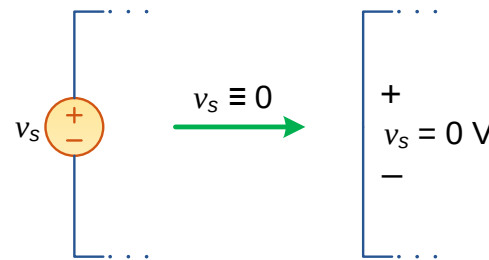
Steps to apply superposition principle:

1. Turn off all **independent** sources except **one** by setting them to **zero**.
2. Find the output (voltage or current in question) using methods covered in Topics 2 and 3.
3. Repeat step 1 for each of the other independent sources.
4. Find the total contribution by **adding** algebraically all the contributions due to the **independent** sources.

Turning off a **current source** means replacing it with an **open circuit**

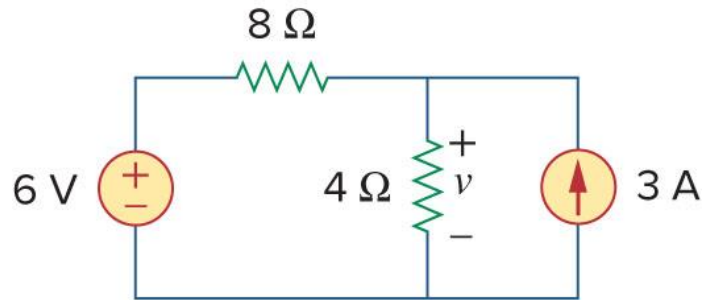


Turning off a **voltage source** means replacing it with a **short circuit**



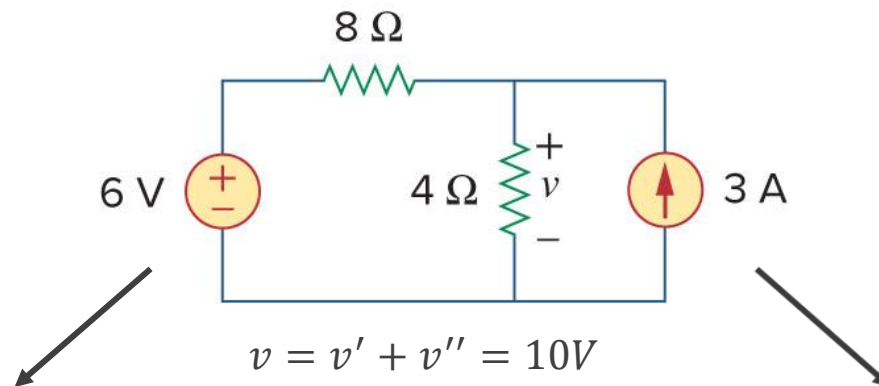
Exercise 1

Using the superposition theorem, find v in the circuit below.

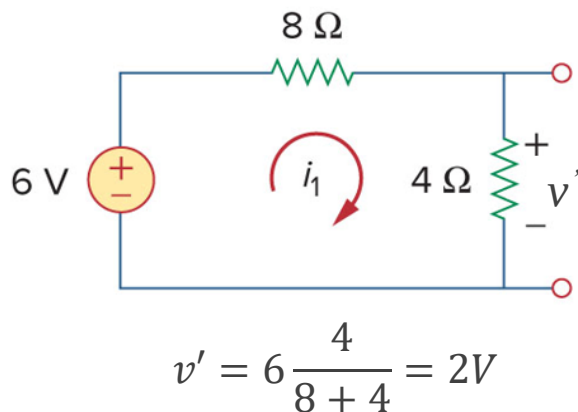


Exercise 1

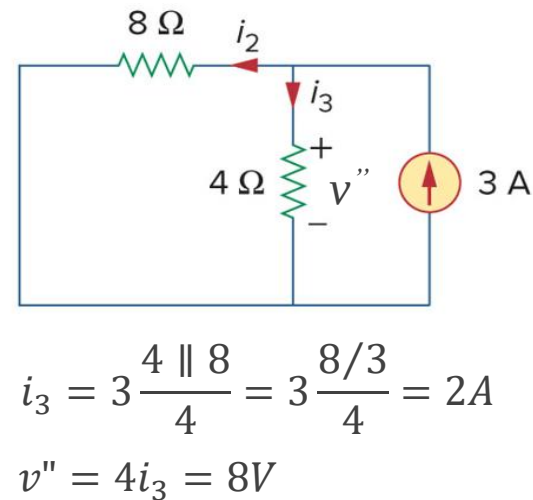
Using the superposition theorem, find v in the circuit below.



Turn off 3A source (open-circuit)



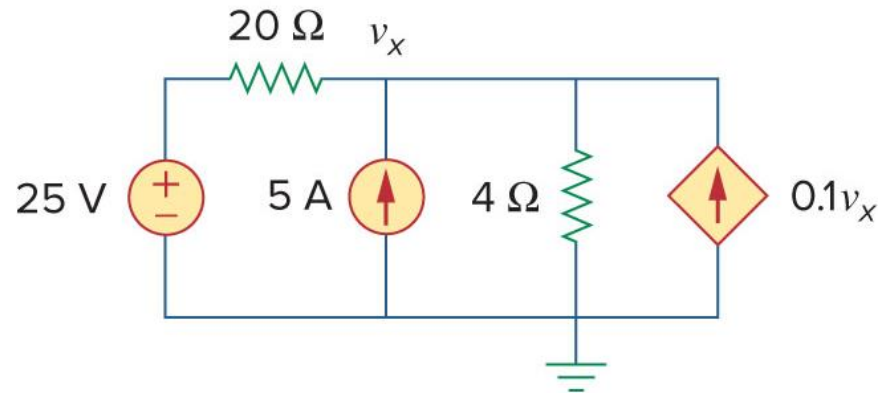
Turn off 6V source (short-circuit)



Exercise 2

Using the superposition theorem, find v_x in the circuit below.

Solution in Exercises_solution_Topic_3.pdf

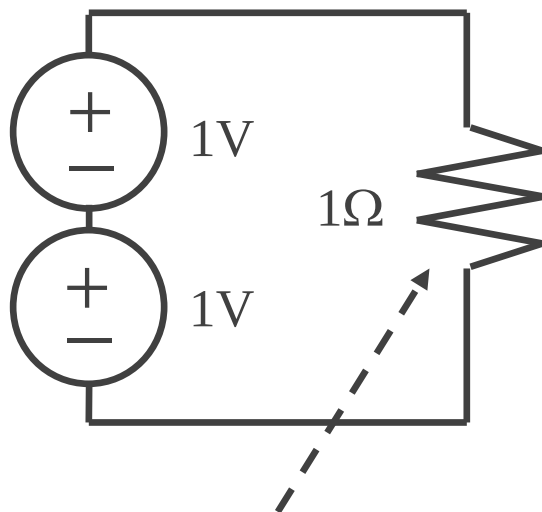


Superposition

Superposition is only applicable to **linear responses**.

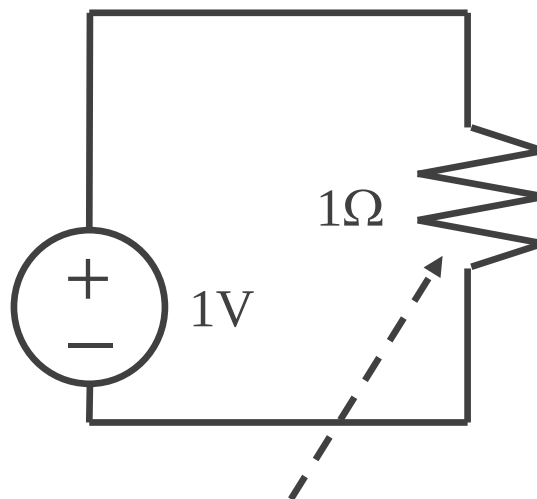
E.g. power in resistors is a nonlinear response.

$$p = vi = i^2 R = \frac{v^2}{R}$$



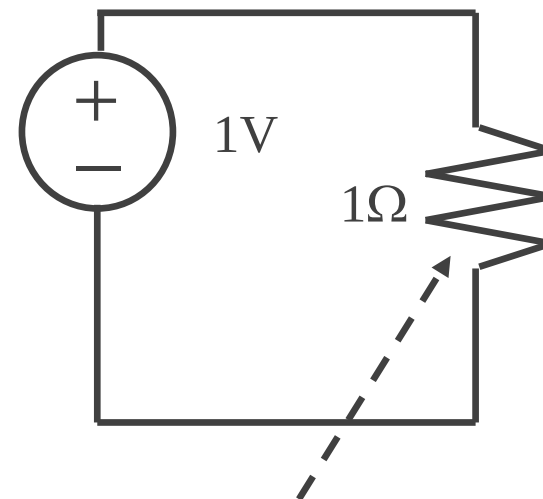
4W delivered
to resistor

≠



1W delivered
to resistor

+

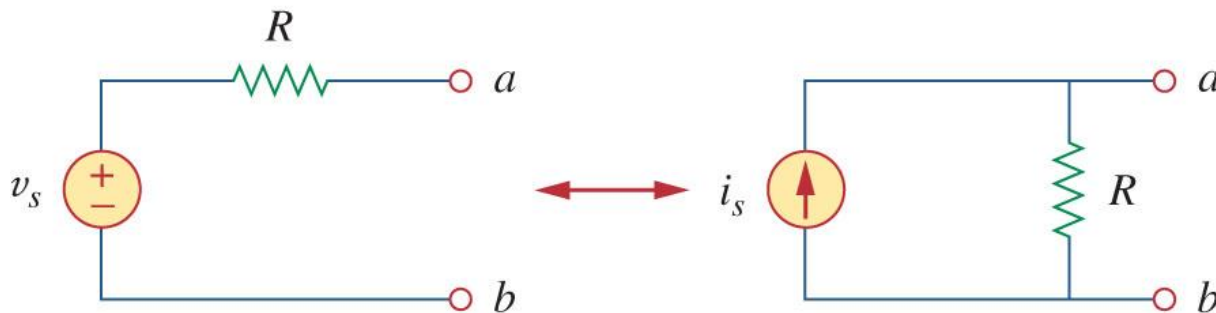


1W delivered
to resistor

Source transformation

- It is possible to transform a source from one form to another. This can be useful for **simplifying** circuits.
- The principle behind this transformation is the concept of **equivalence**, where two circuits are called equivalent if their **v - i characteristics** are **identical** at the terminal.
- The **direction** of current and **polarity** of voltage should follow the **passive sign convention**.

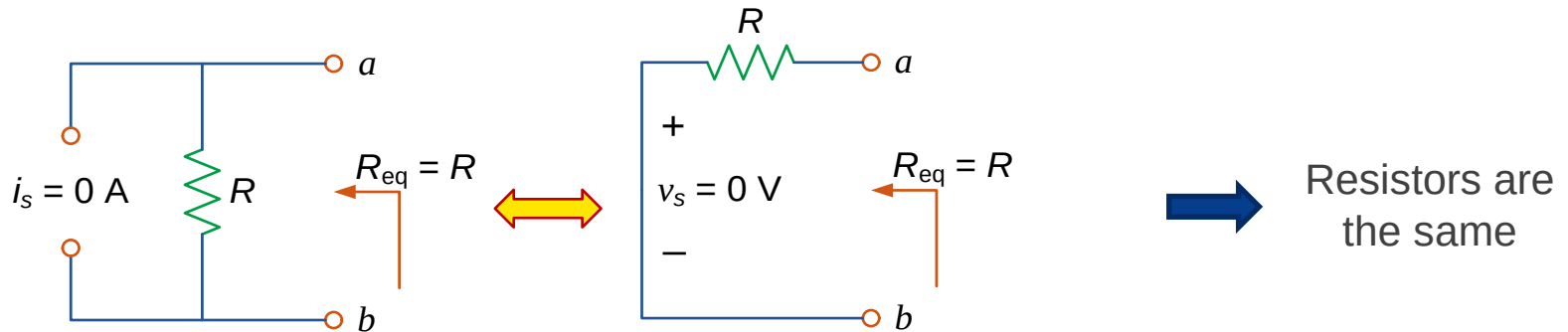
A **source transformation** is the process of **replacing** a voltage source v_s in **series** with a resistor R by a current source i_s in **parallel** with a resistor R or vice versa



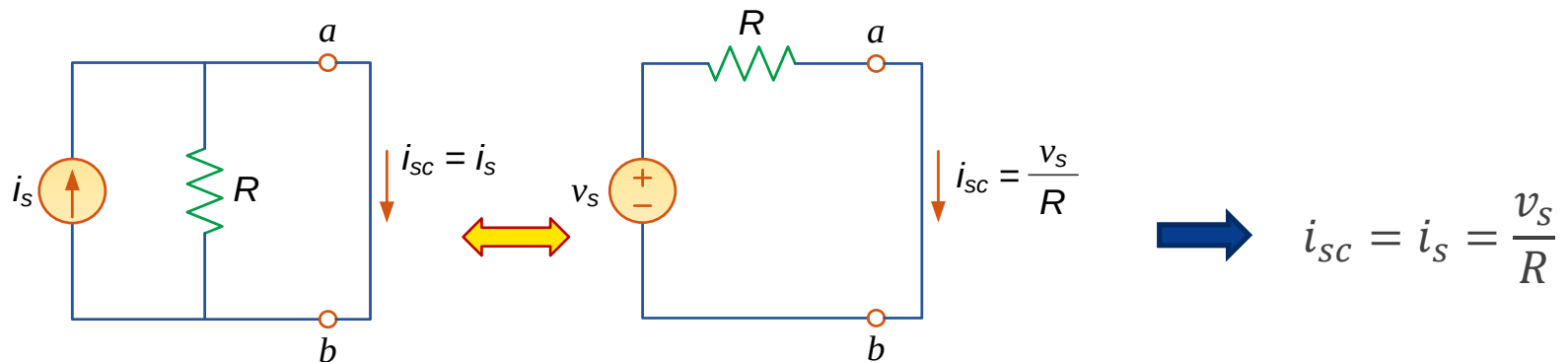
Source transformation

Terminal equivalency

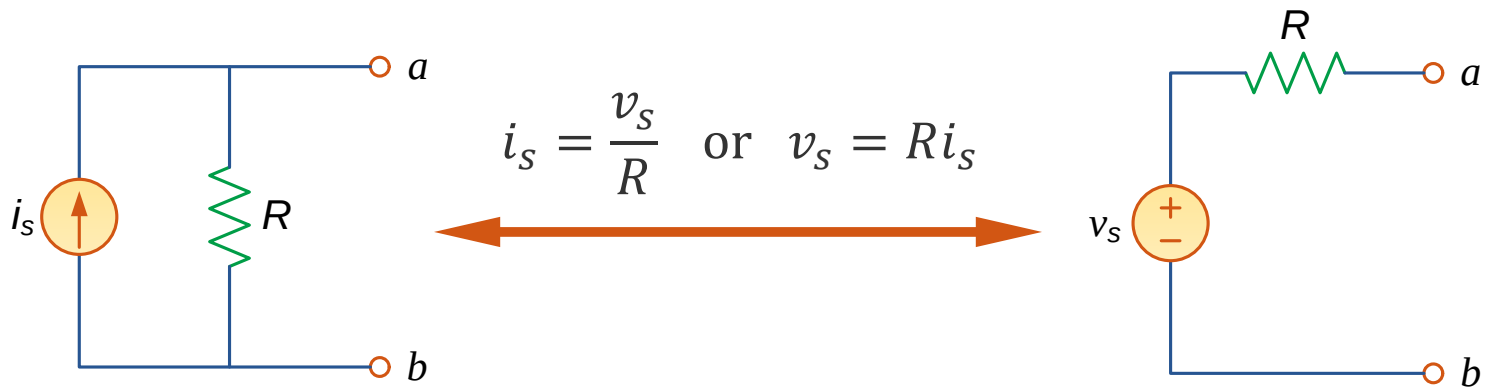
- If we turn off the sources, the **equivalent resistance** (R_{eq}) at terminal a - b must be the **same**.



- Short-circuit current** (i_{sc}) flowing from a to b must be the **same**.

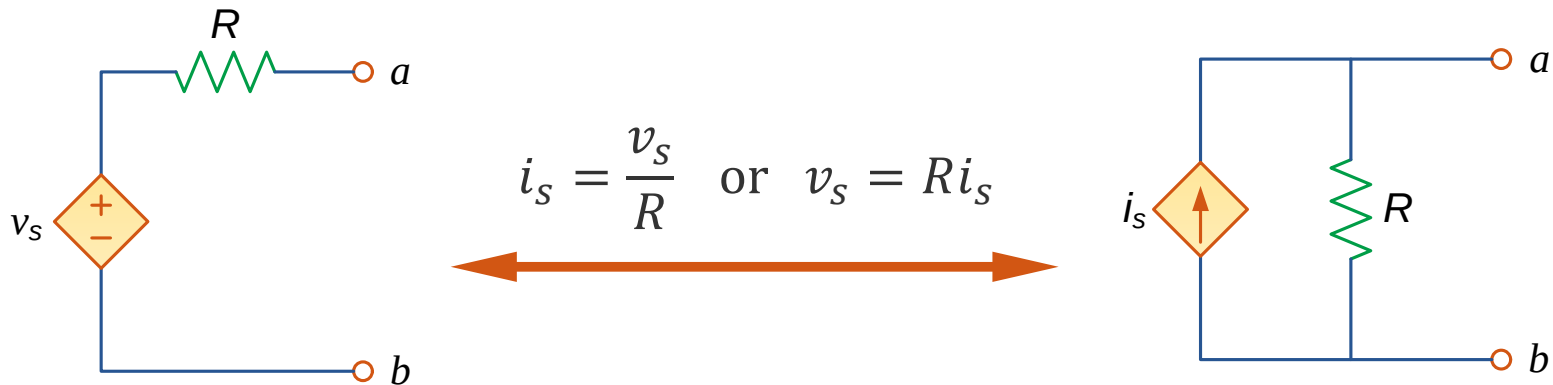


Source transformation



Source transformation

- Source transformation can also be applied to **dependent** sources, provided that the dependent **variable** is handled **carefully**.



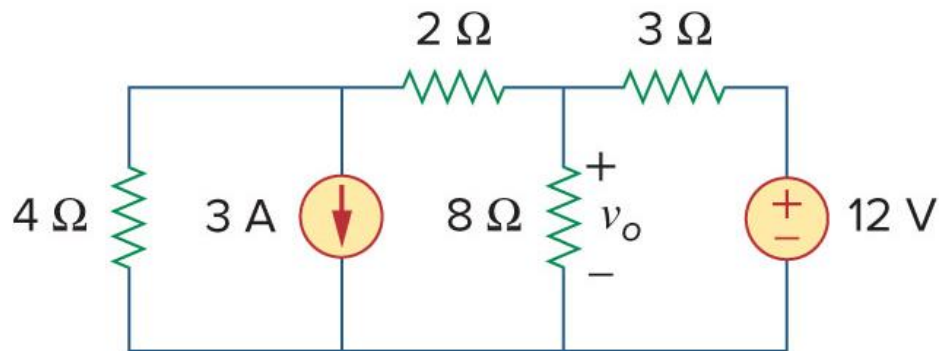
Some limitations:

- Source transformation is **not** possible when $R = 0$ for an **ideal voltage source** (for a practical voltage source $R \neq 0$)
- Source transformation is **not** possible when $R = \infty$ for an **ideal current source** (for a practical current source $R \neq \infty$)

Exercise 3

Use source transformation to find v_o in the circuit below.

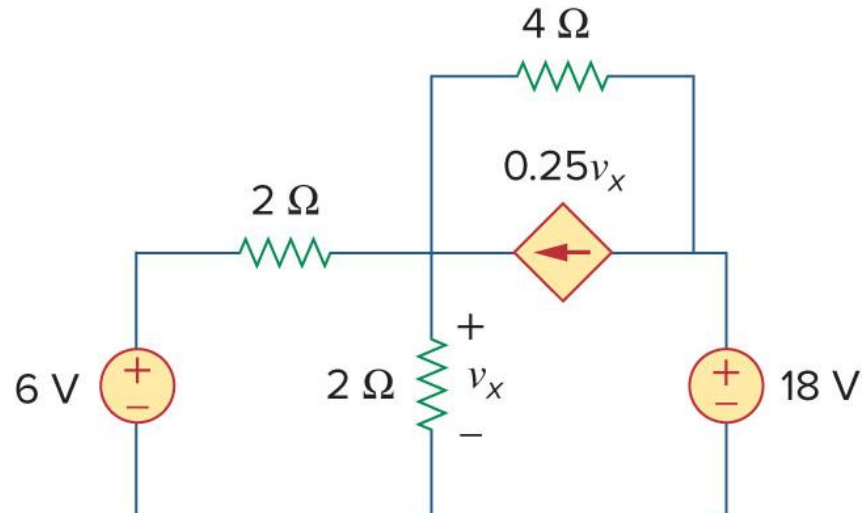
Solution in Exercises_solution_Topic_3.pdf



Exercise 4

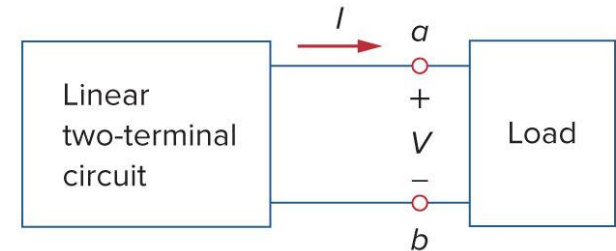
Use source transformation to find v_x in the circuit below.

Solution in Exercises_solution_Topic_3.pdf

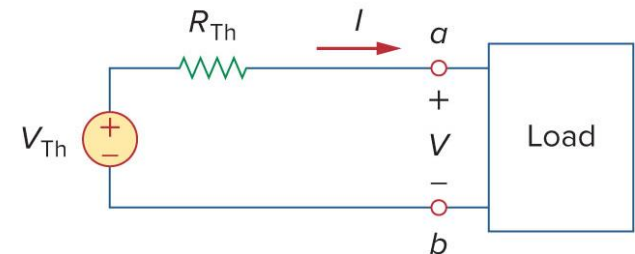


Thevenin's theorem

- In many circuits, one element can be variable (e.g. house hold appliances consuming different powers – hairdryer, fridge, etc.). This variable element is called the **load**.
- Every time the load changes, the circuit would have to be analyzed again.
- Thevenin's theorem provides a technique to simplify the analysis by **replacing the fixed part** of the circuit with an equivalent one known as **Thevenin equivalent circuit**.



(a)

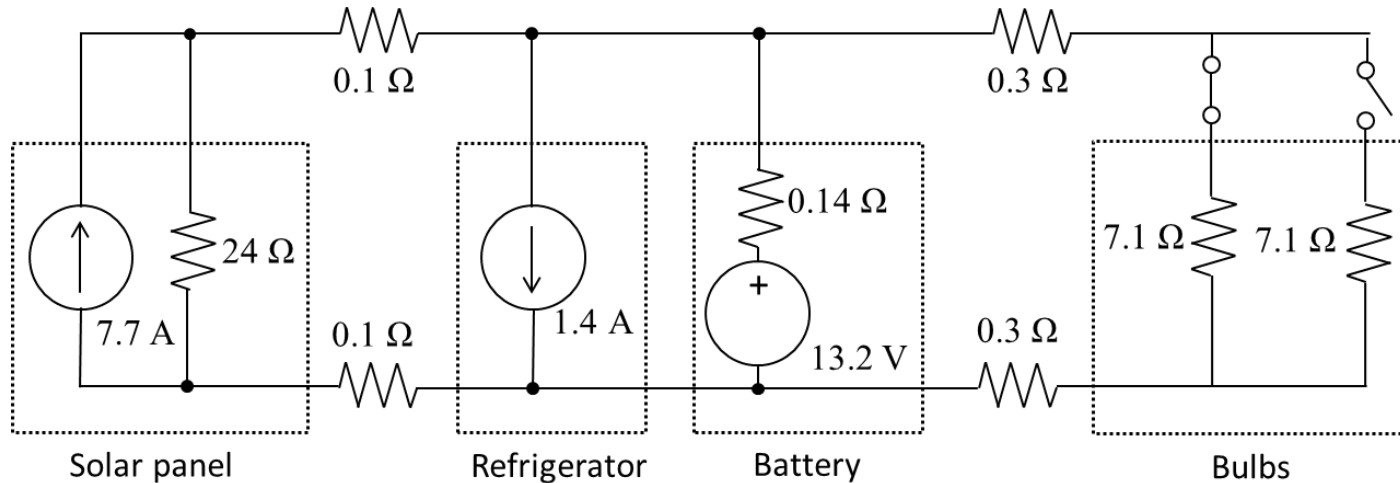


(b)

Thevenin's theorem states that a **linear** two-terminal circuit (Fig. (a)) can be replaced by an **equivalent circuit** consisting of a **voltage source V_{Th}** in **series** with a **resistor R_{Th}** (Fig. (b))

Simulation

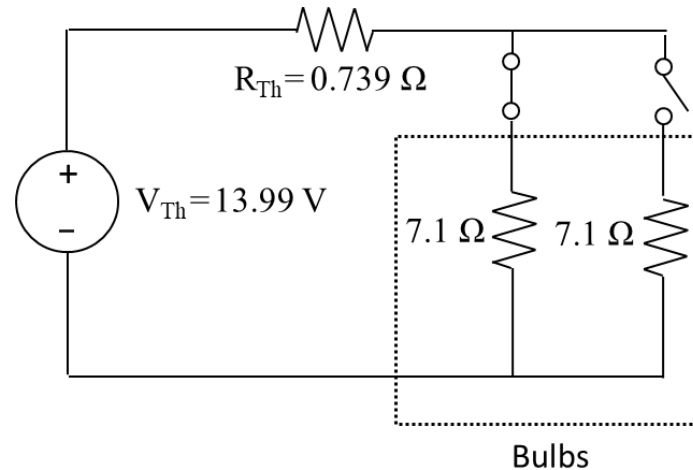
The circuit below corresponds to a solar-powered caravan. What happens with the current in the bulbs every time we open or close a switch?



<http://tinyurl.com/y64aqyk2>

Simulation

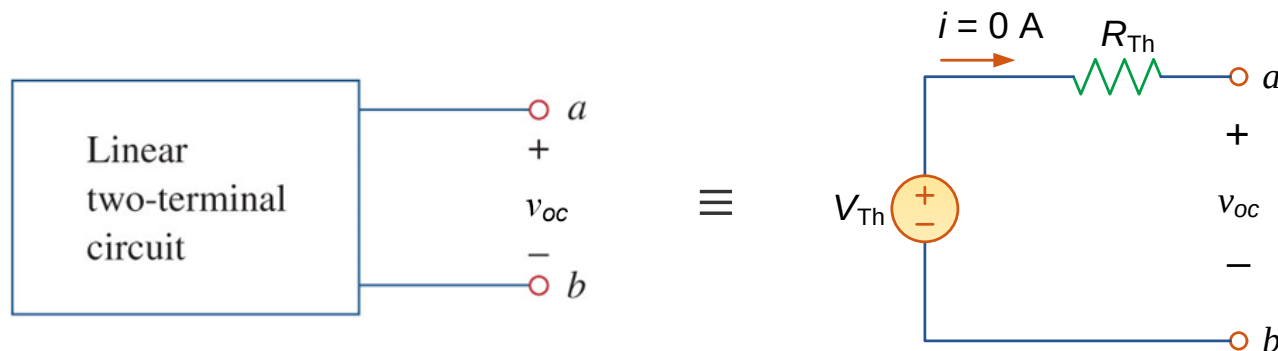
Let's replace the solar powered caravan with its corresponding Thevenin equivalent... What happens with the current in the bulbs every time we open or close a switch?



<http://tinyurl.com/y6fb695h>

Thevenin's theorem

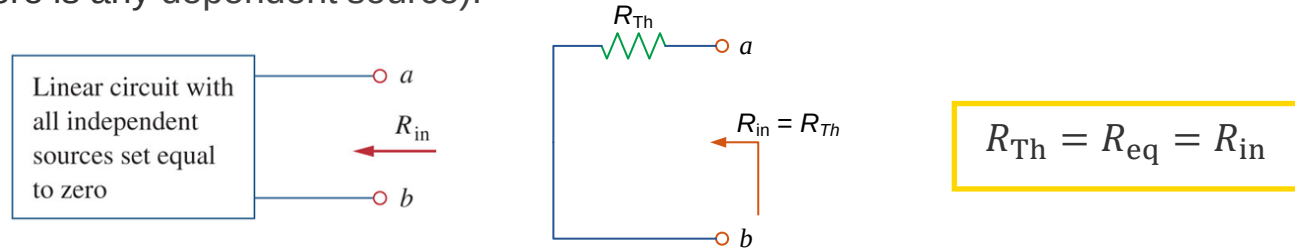
- The voltage source's value, known as **Thevenin voltage** V_{Th} , is equal to the **open-circuit voltage** at the terminals.



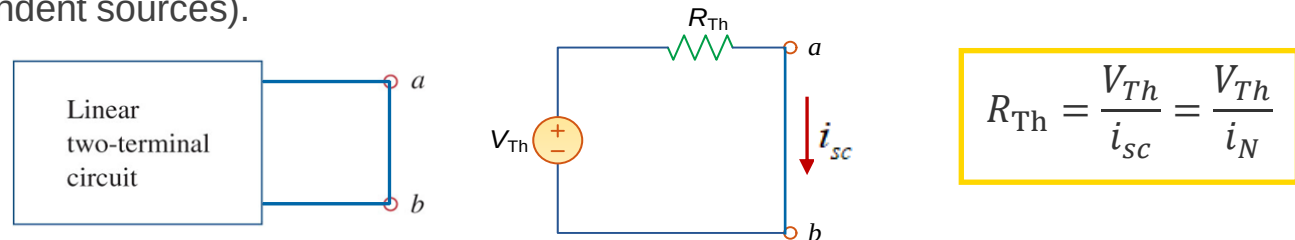
$$V_{Th} = v_{oc}$$

Thevenin's theorem

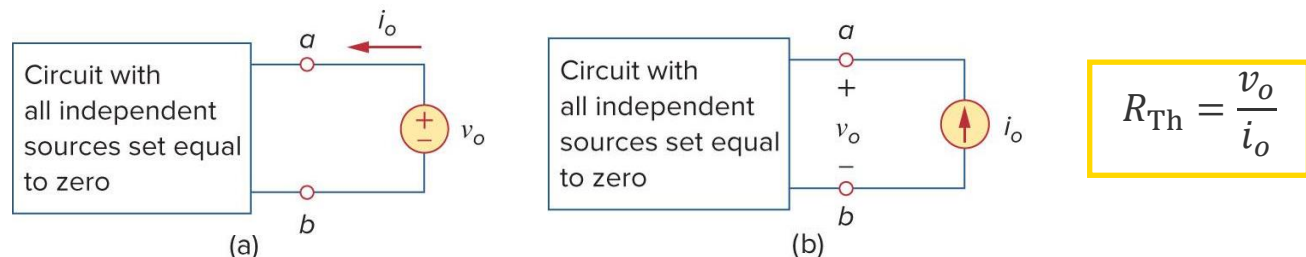
- R_{Th} can be obtained using different methods:
 1. Input **resistance** measured at the terminal pair when **all independent sources are turned off** (not valid if there is any dependent source).



2. Ratio of the open-circuit voltage to the short-circuit current at the terminal pair (not valid if there are only dependent sources).

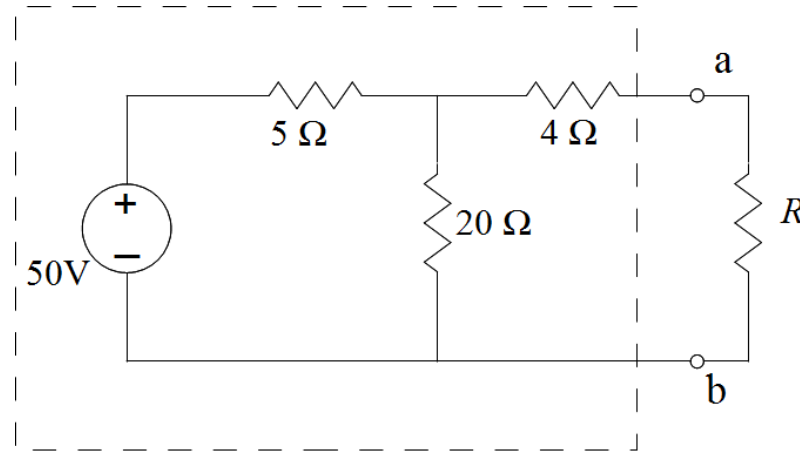


3. Turn off independent sources and (a) attach a voltage source v_o to the terminals a - b and find the resulting current i_o , or (b) attach a current source i_o and find the resulting voltage v_o . For simplicity: $v_o = 1$ V or $i_o = 1$ A (valid always).



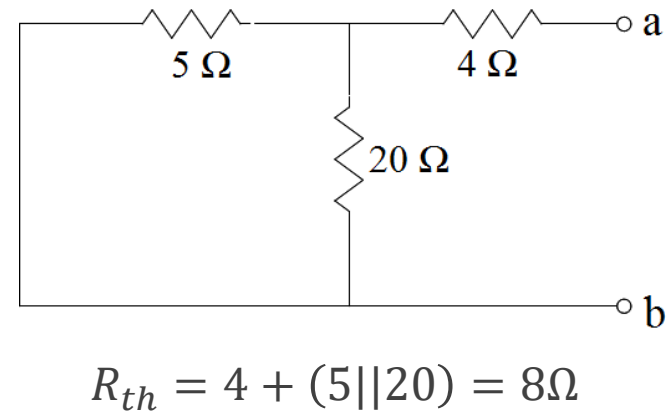
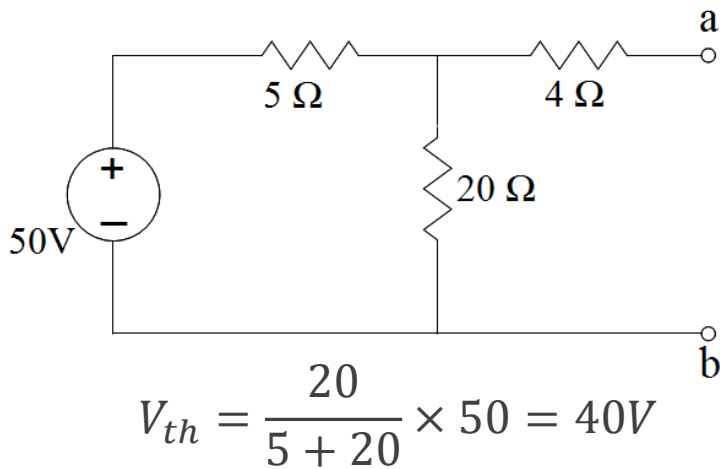
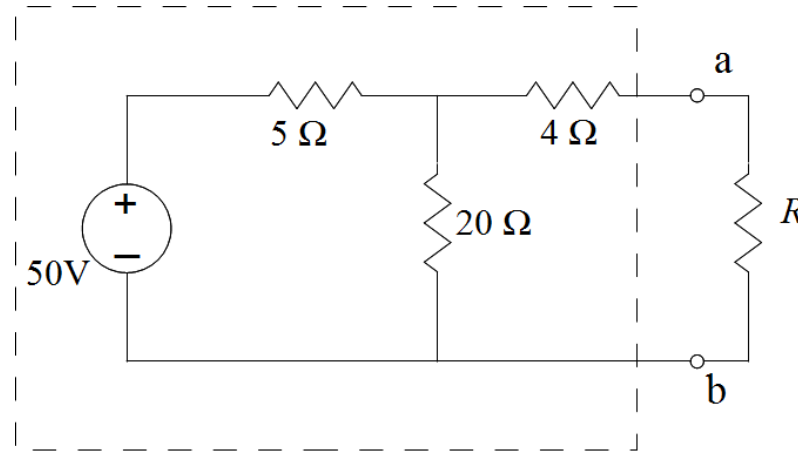
Exercise 5

Replace the circuit in the box with its Thevenin equivalent circuit.



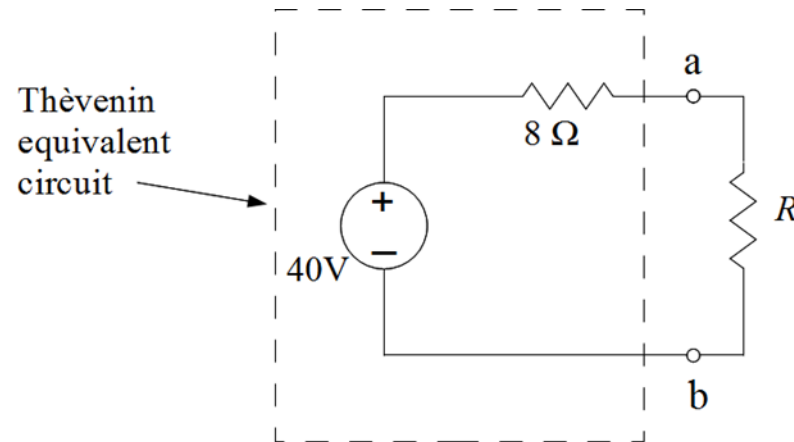
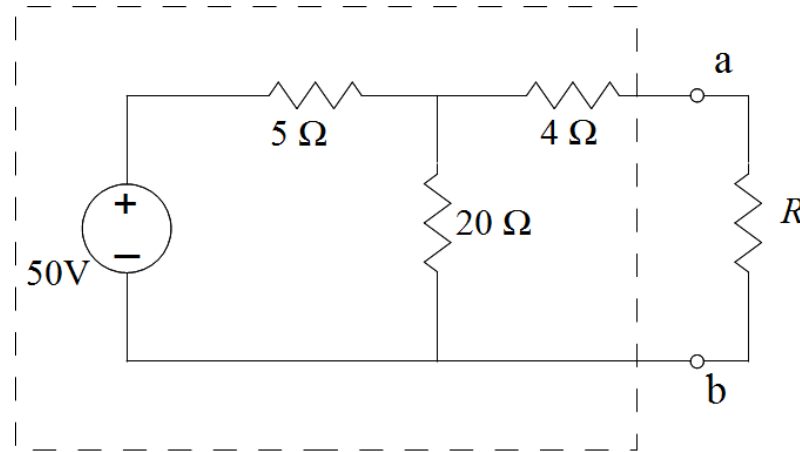
Exercise 5

Replace the circuit in the box with its Thevenin equivalent circuit.



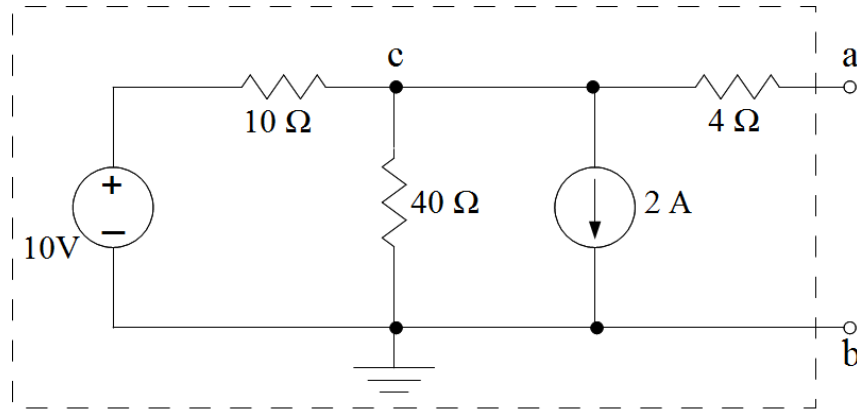
Exercise 5

Replace the circuit in the box with its Thevenin equivalent circuit.



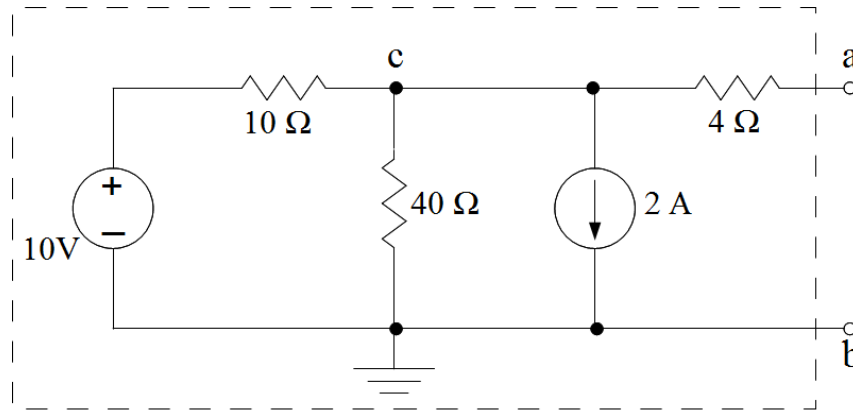
Exercise 6

Calculate the Thevenin equivalent of the circuit, as seen from terminals a-b.



Exercise 6

Calculate the Thevenin equivalent of the circuit, as seen from terminals a-b.

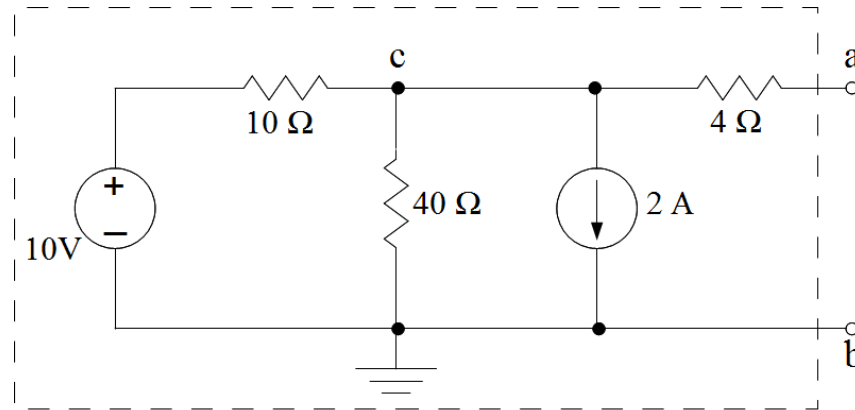


- For open-circuit voltage use nodal analysis:

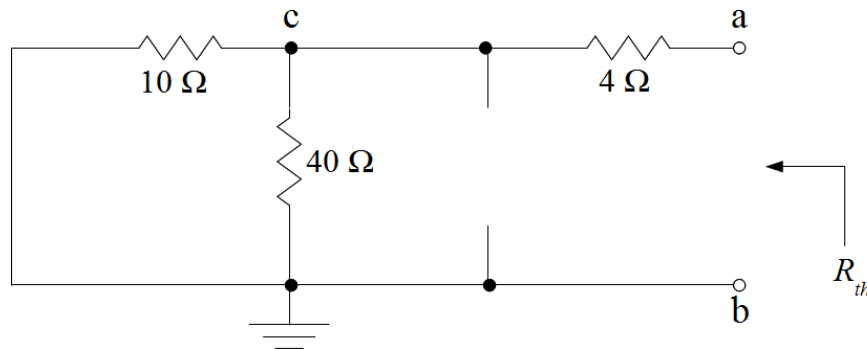
$$\frac{10 - v_c}{10} - \frac{v_c}{40} - 2 = 0 \Rightarrow V_{th} = v_{oc} = -8V$$

Exercise 6

Calculate the Thevenin equivalent of the circuit, as seen from terminals a-b.



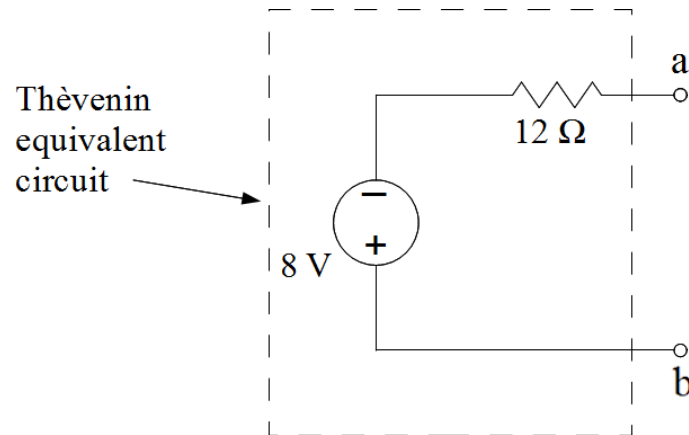
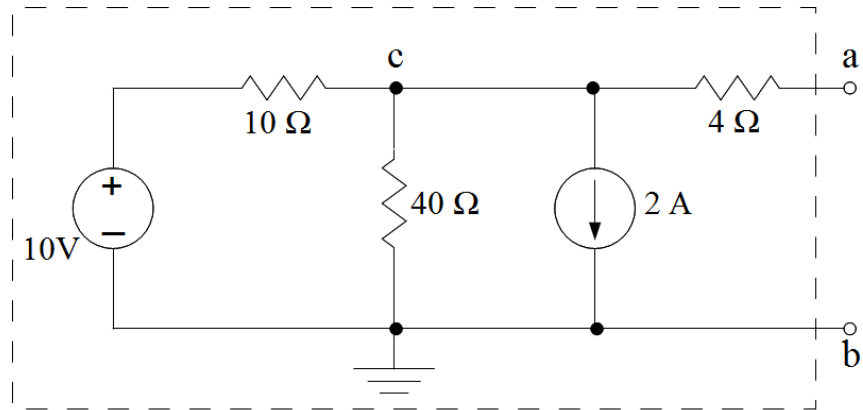
- For R_{th} , turn off independent sources and calculate equivalent resistance as seen from terminals a-b:



$$R_{th} = 4 + (10||40) = 12\Omega$$

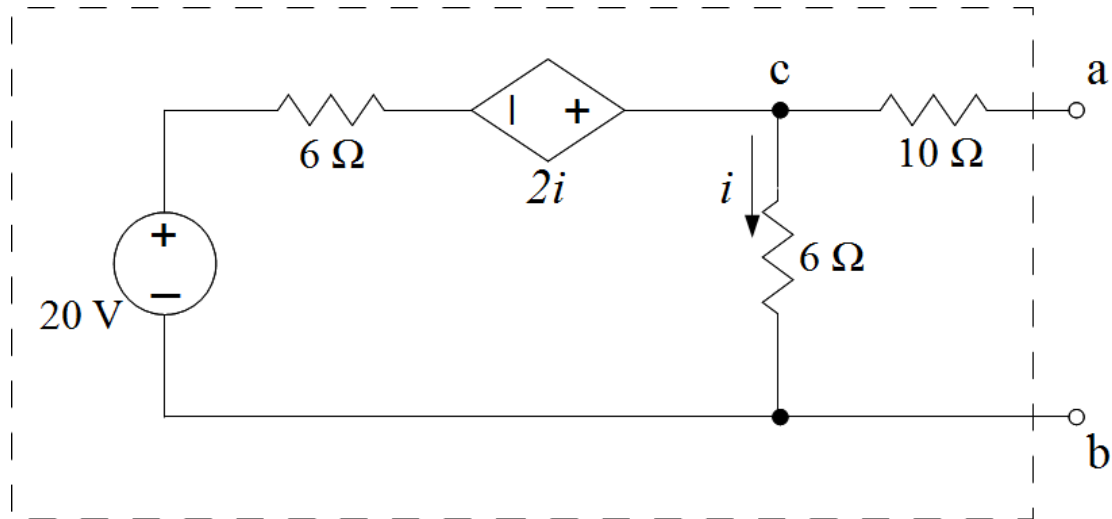
Exercise 6

Calculate the Thevenin equivalent of the circuit, as seen from terminals a-b.



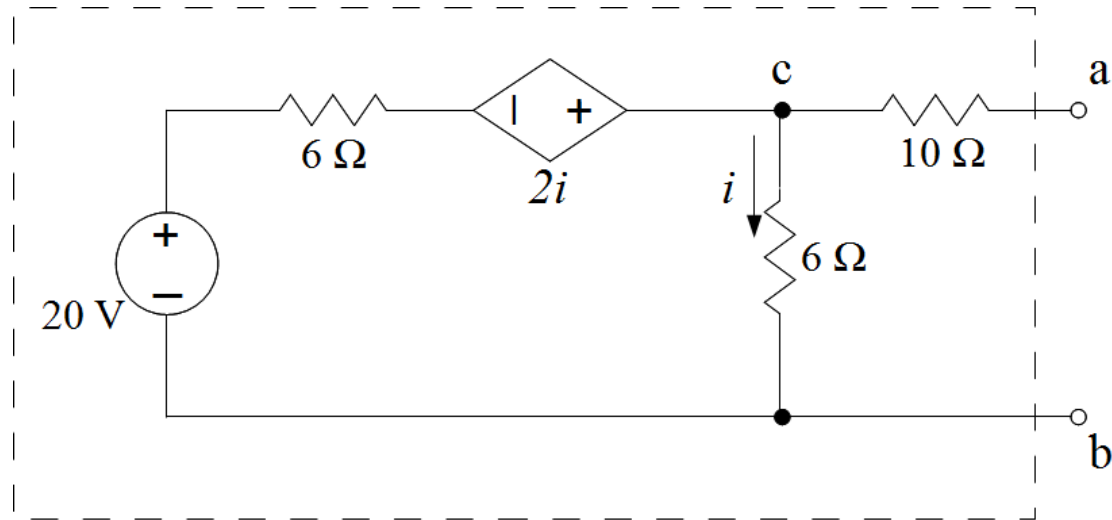
Exercise 7

Calculate the Thevenin equivalent of the circuit, as seen from terminals a-b.



Exercise 7

Calculate the Thevenin equivalent of the circuit, as seen from terminals a-b.

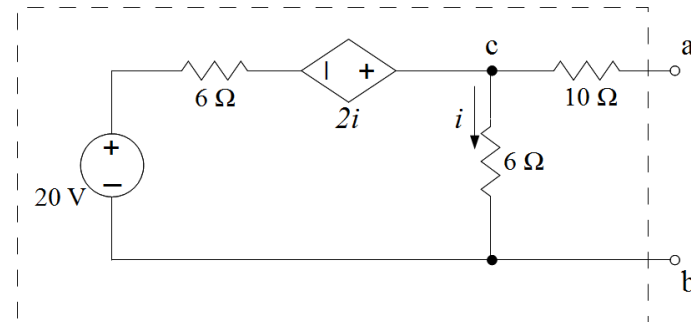


- For open-circuit voltage V_{th} , voltage across terminals a-b is equal to voltage across terminals c-b. Apply KVL:

$$-20 + 6i - 2i + 6i = 0 \quad \Rightarrow \quad i = \frac{20}{10} = 2A \quad \Rightarrow \quad V_{Th} = 2 \times 6 = 12V$$

Exercise 7

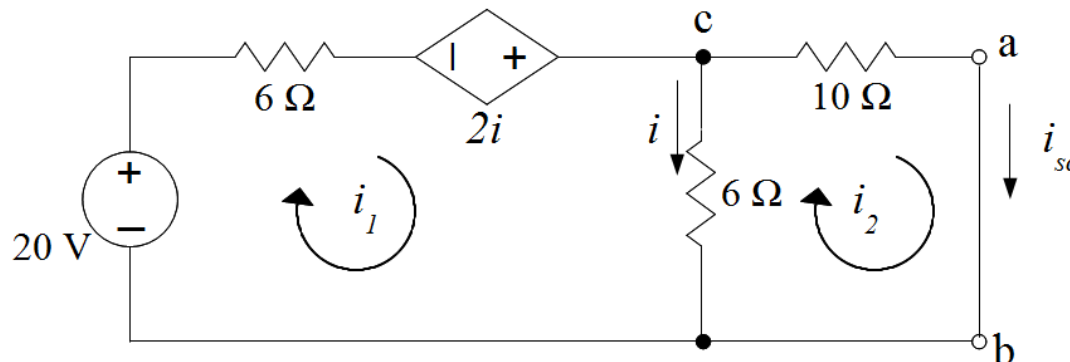
Calculate the Thevenin equivalent of the circuit, as seen from terminals a-b.



- To determine i_{sc} , short-circuit output terminals, and use mesh equations:

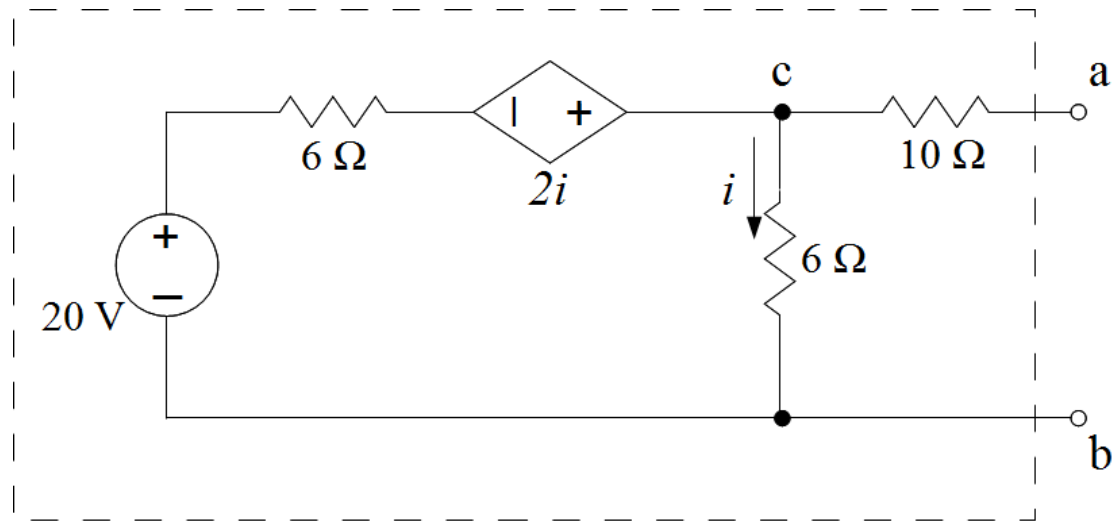
$$\begin{cases} -20 + 6i_1 - 2i + 6(i_1 - i_2) = 0 \\ -6(i_1 - i_2) + 10i_2 = 0 \\ i = i_1 - i_2 \end{cases}$$

$$i_{sc} = i_2 = \frac{30}{34} A \Rightarrow R_{th} = \frac{V_{th}}{i_{sc}} = \frac{12}{30/34} = 13.6\Omega$$

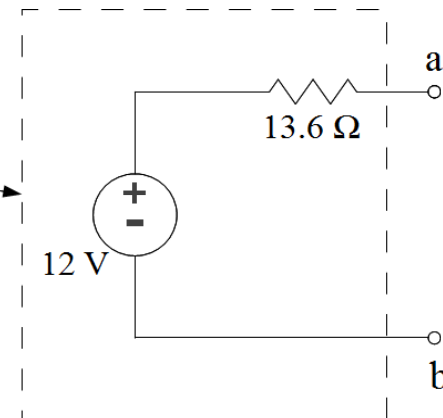


Exercise 7

Calculate the Thevenin equivalent of the circuit, as seen from terminals a-b.



Thevenin
equivalent
circuit



Thevenin's theorem

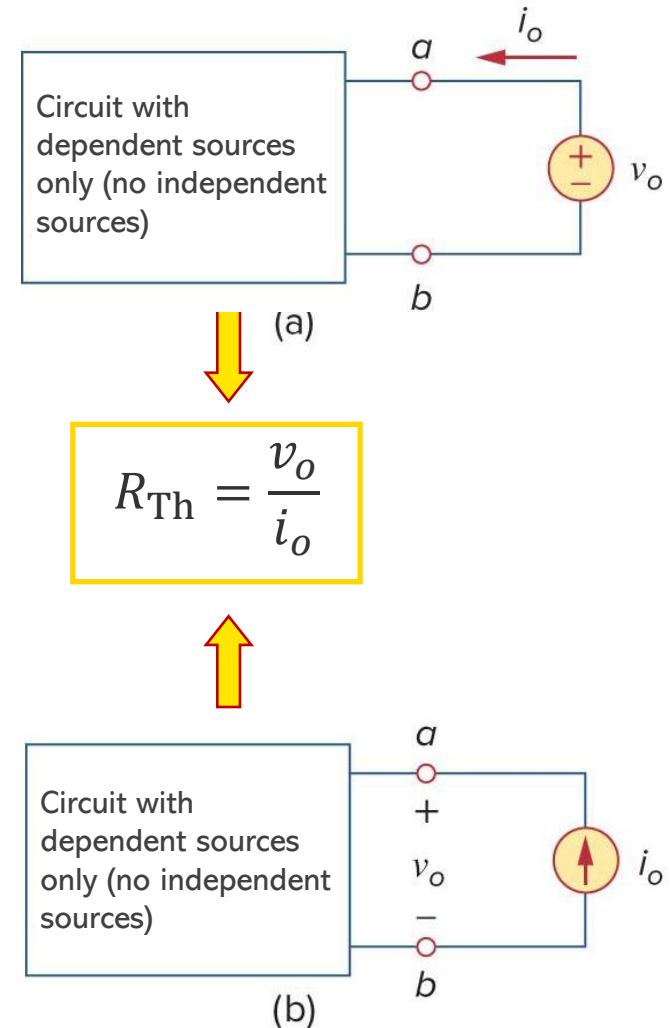
When we have a circuit with **dependent sources but not independent ones**, we have a “dead network”, i.e. all voltages and currents are equal to zero.

- Thevenin voltage V_{Th} :

$$V_{Th} = v_{oc} = 0$$

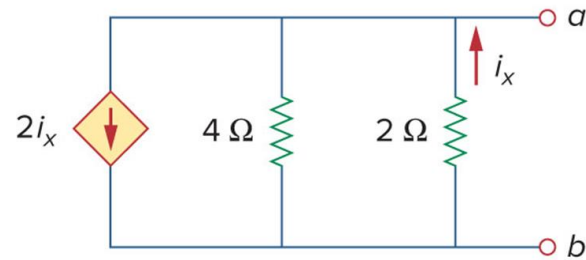
- Thevenin resistance R_{Th} :

- All currents and voltages in the circuit are equal to zero, including i_{sc} .
- The only possible method to calculate R_{Th} in this case is method 3, i.e. attach a **voltage source** v_o to the terminals **a-b** and find the resulting **current** i_o (Fig. (a)), or attach a **current source** i_o and find the resulting **voltage** v_o (Fig. (b)).
- Note: In this case it is possible for R_{Th} to be negative ($R_{Th} < 0$). This implies that the circuit is supplying power.



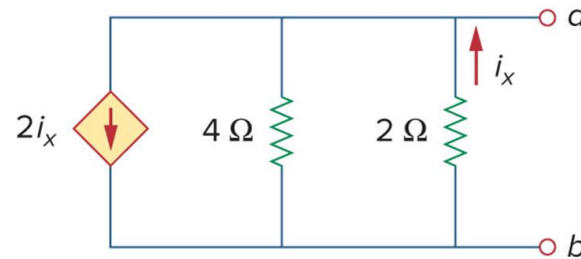
Exercise 8

Find the Thevenin equivalent of the circuit below at the terminals a - b .

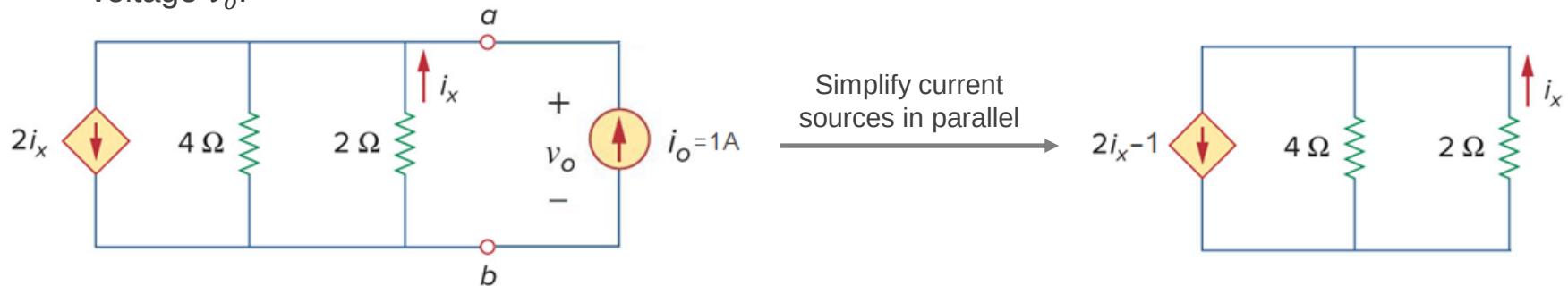


Exercise 8

Find the Thevenin equivalent of the circuit below at the terminals a - b .



- $V_{Th} = v_{oc} = 0$ (since there are no independent sources in the circuit).
- To calculate R_{Th} we attach a current source i_o ($i_o = 1$ A) to the terminals and find the resulting voltage v_o .



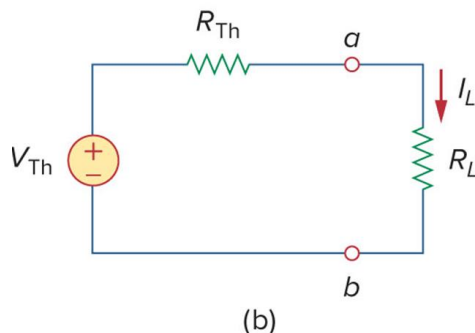
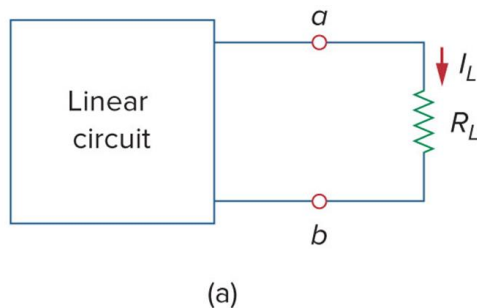
Use current division to find i_x :
$$i_x = (2i_x - 1) \frac{4 \parallel 2}{2} = (2i_x - 1) \frac{\frac{1}{\frac{1}{4} + \frac{1}{2}}}{2} = (2i_x - 1) \frac{4}{4+2} = \frac{2}{3}(2i_x - 1)$$

$$i_x - \frac{4}{3}i_x + \frac{2}{3} = 0; \quad 3i_x - 4i_x + 2 = 0 \rightarrow i_x = 2A$$

$$v_o = -2i_x = -4V \rightarrow R_{Th} = \frac{v_o}{i_o} = \frac{-4}{1} = -4\Omega$$

Thevenin's theorem

- Thevenin's theorem is a powerful technique in circuit analysis with variable loads.
- It allows to **simplify** a large linear circuit.
- The **equivalent circuit** behaves externally **exactly the same** as the original circuit.
- The current through the **load** R_L (**load current** I_L) and the voltage across the load (**load voltage** V_L) is obtained using simple voltage division or KVL/KCL.



$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$
$$V_L = \frac{R_L}{R_{Th} + R_L} V_{Th}$$

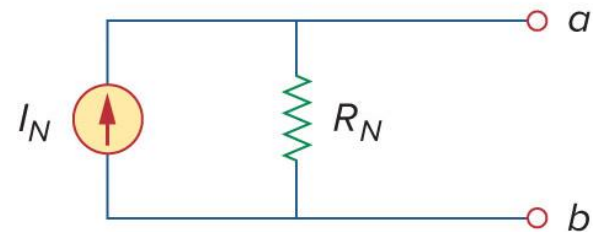
Norton's theorem

- Norton's theorem is the **dual** form of Thevenin's theorem.
- It provides a similar technique to **simplify** the analysis by **replacing a linear circuit** with an equivalent one known as **Norton's equivalent circuit**.

Norton's theorem states that a **linear** two-terminal circuit (Fig. (a)) can be replaced by an **equivalent circuit** consisting of a **current source** I_N in **parallel** with a **resistor** R_N (Fig. (b)).



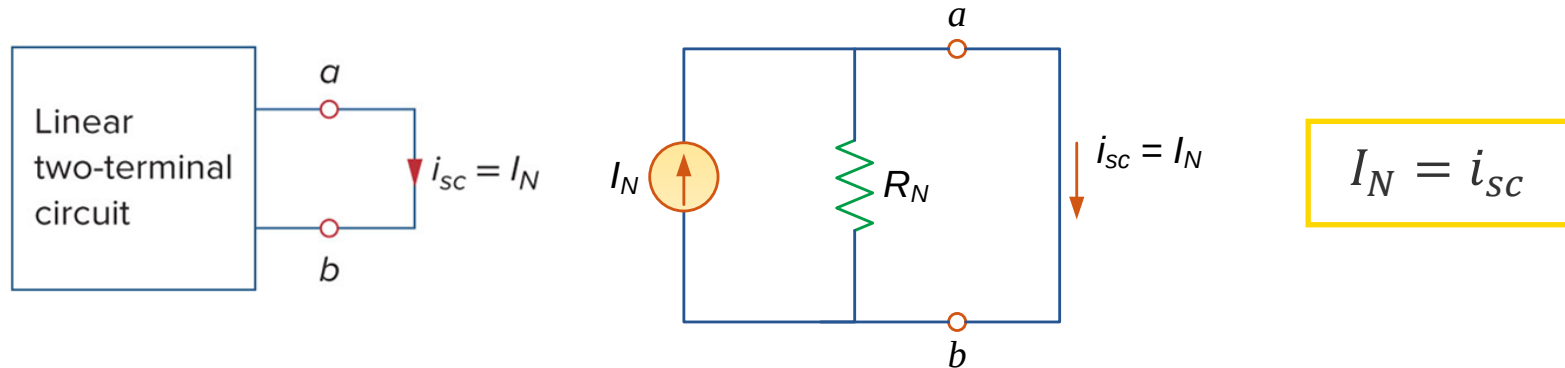
(a)



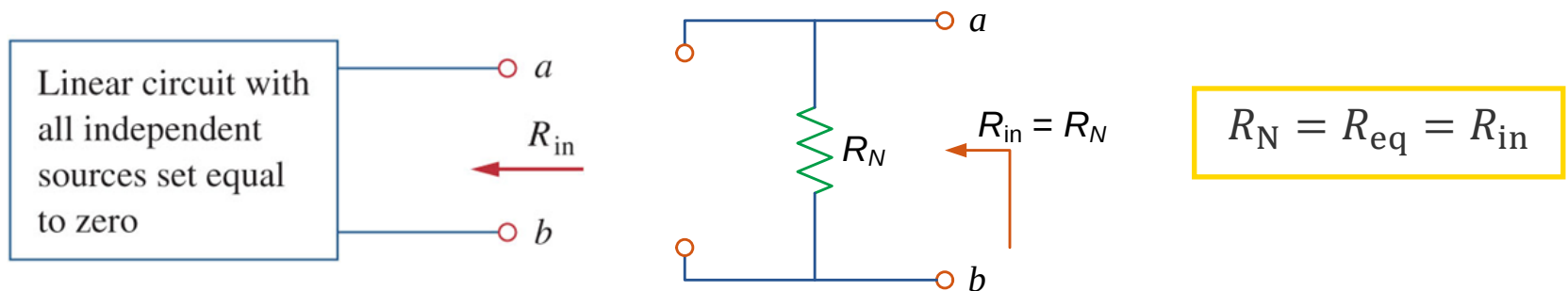
(b)

Norton's theorem

- The current source's value, known as Norton current I_N , is equal to the short-circuit current at the terminals.

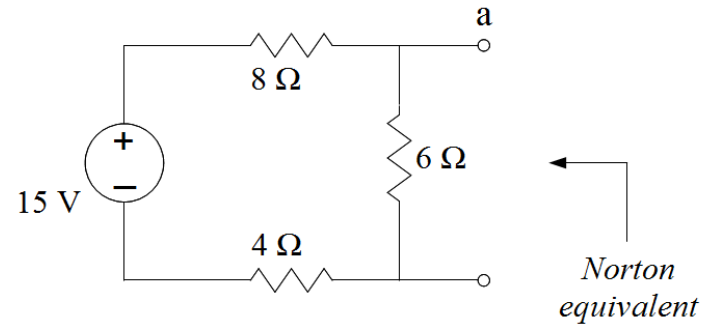


- The **Norton resistance R_N** is **same** as the **Thevenin resistance R_{Th}** , which is the **input resistance** measured at the terminals when **all independent sources are turned off**.



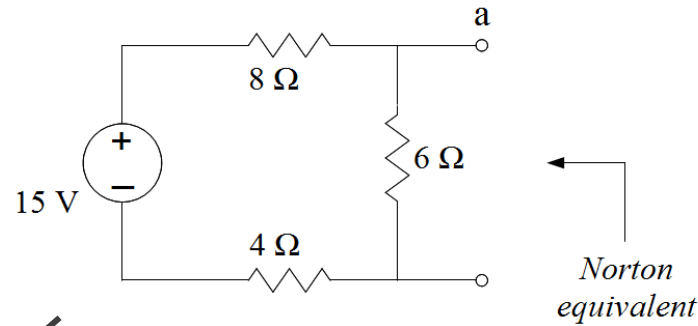
Exercise 9

Calculate the Norton equivalent of the following circuit:

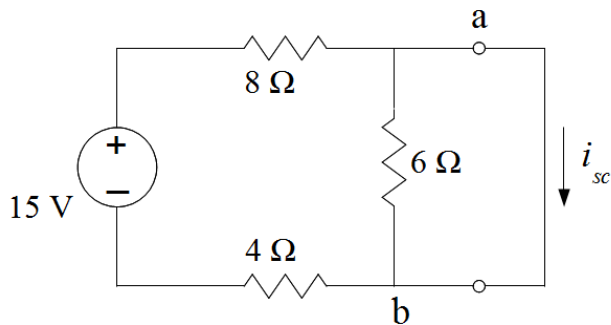


Exercise 9

Calculate the Norton equivalent of the following circuit:

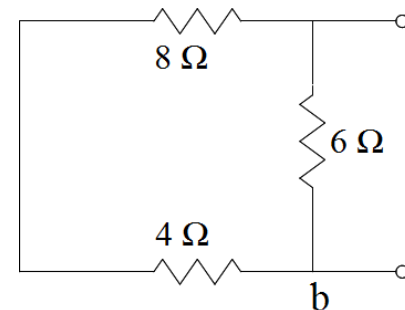


Norton current (i_{sc})



$$i_N = i_{sc} = \frac{15}{8 + 4} = 1.25 \text{ A}$$

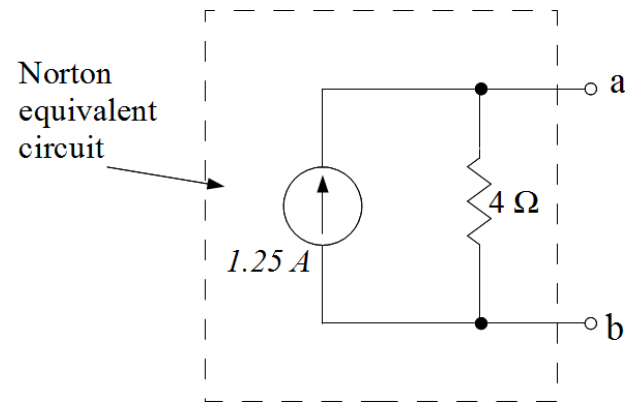
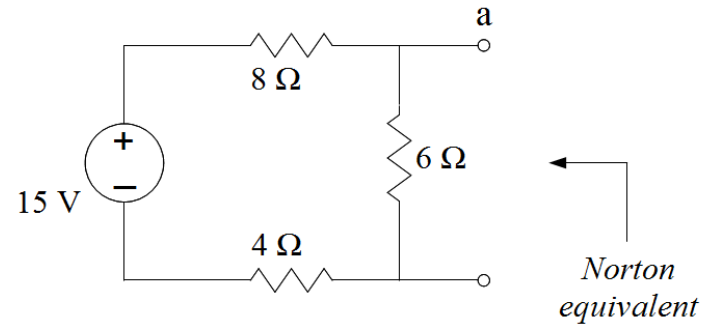
Norton resistance (R_N)



$$R_N = 6 \parallel (8 + 4) = 4 \Omega$$

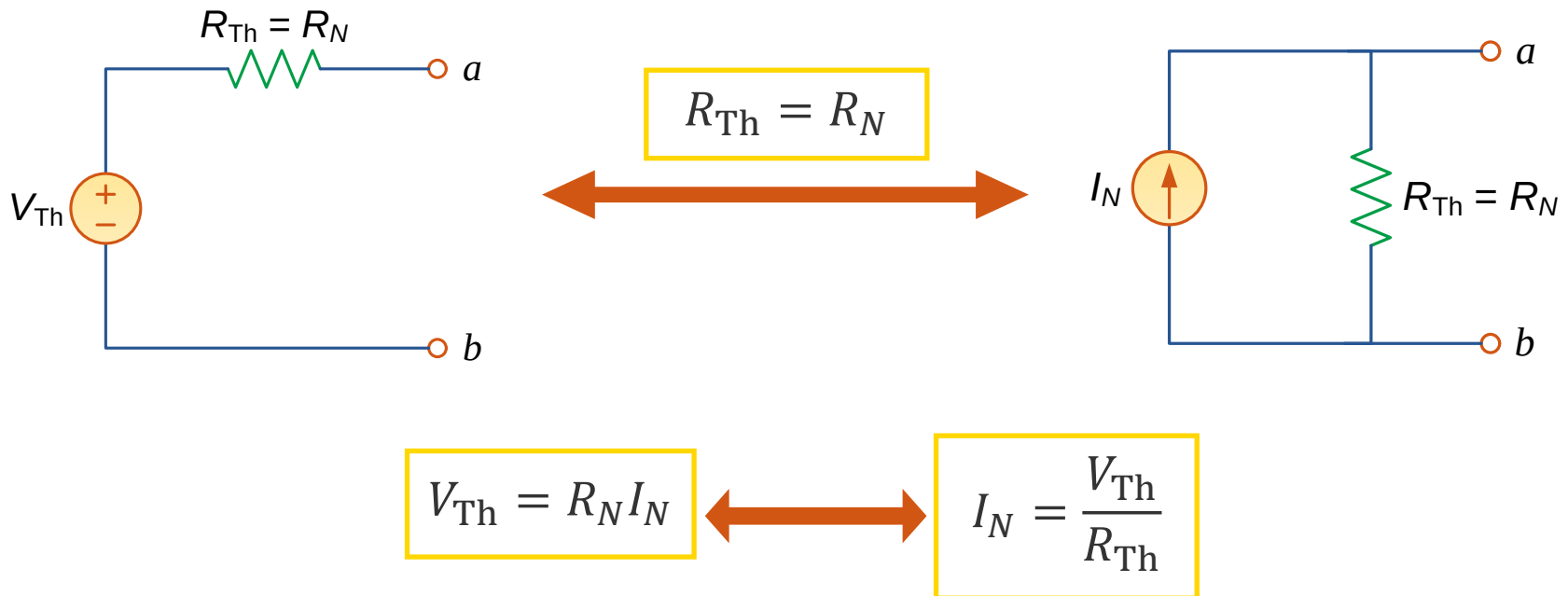
Exercise 9

Calculate the Norton equivalent of the following circuit:



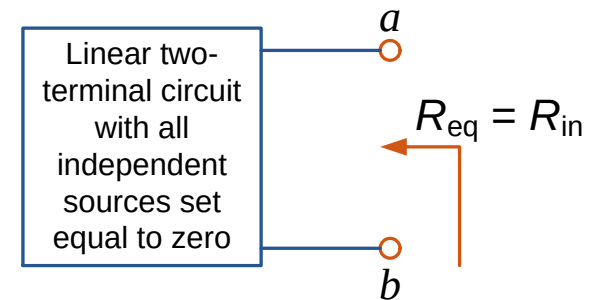
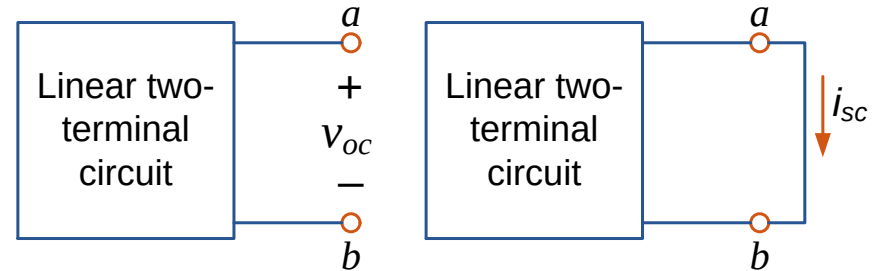
Thevenin-Norton transformation

- Thevenin's and Norton's theorems are **related** to each other through **source transformation**.
- Thevenin and Norton resistances are **exactly the same** and equal to **input resistance** measured from the terminals of the circuit.



Thevenin-Norton transformation

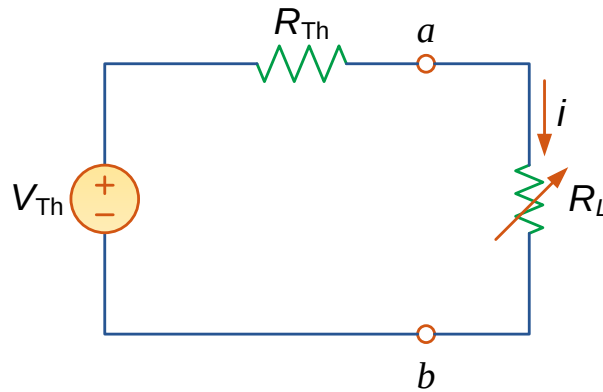
- **Dependent** sources in Norton equivalent circuit are handled in **the same way** as Thevenin equivalent circuit.
- Since V_{Th} , I_N and $R_{Th} = R_N$ are related based on source transformation, finding the Thevenin or Norton equivalent circuit requires **two** of the following:
 1. The **open-circuit voltage** v_{oc} across terminals a and b .
 2. The **short-circuit current** i_{sc} at terminals a and b .
 3. The **equivalent or input resistance** $R_{eq} = R_{in}$ at terminals a and b when all **independent** sources are **turned off**.



$$\begin{aligned} V_{Th} &= v_{oc} \\ I_N &= i_{sc} \\ R_{Th} &= \frac{v_{oc}}{i_{sc}} = R_N \end{aligned}$$

Maximum power transfer

- In many practical applications, a circuit is designed to provide **maximum power** to a **load**.
- Unlike ideal sources, **internal resistance** of real sources **restricts** the amount of power that can be transferred to a load.
- The power consumption of that internal resistance is known as **power loss**.
- Using the **Thevenin equivalent circuit** we can find the **maximum power** that can be transferred to a load based on **fixed** V_{Th} and R_{Th} and **variable** R_L .



$$p(R_L) = R_L i^2 = R_L \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2$$

Maximum power transfer

- To find for which value of R_L the power would be maximum, differentiate p with respect to R_L and set the result to zero.

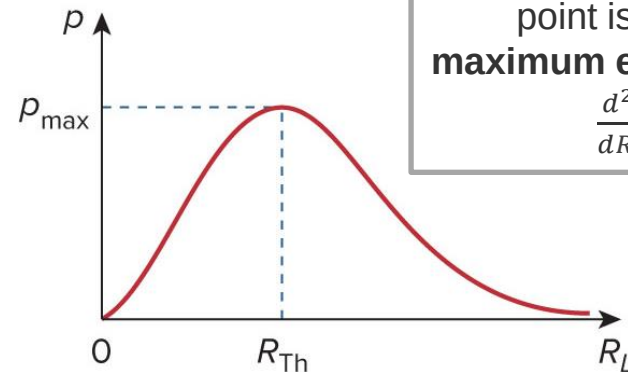
$$\begin{aligned}\frac{dp}{dR_L} &= V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right] \\ &= V_{Th}^2 \left[\frac{(R_{Th} + R_L - 2R_L)}{(R_{Th} + R_L)^3} \right] = 0 \quad \Rightarrow \quad (R_{Th} + R_L - 2R_L) = 0\end{aligned}$$

- Maximum power** is transferred to the load when the **load resistance R_L** is **equal** to the **Thevenin resistance R_{Th}** as seen from the load terminals

$R_L = R_{Th}$

➔

$p_{\max} = \frac{V_{Th}^2}{4R_{Th}}$
 $v_L = \frac{V_{Th}}{2}$



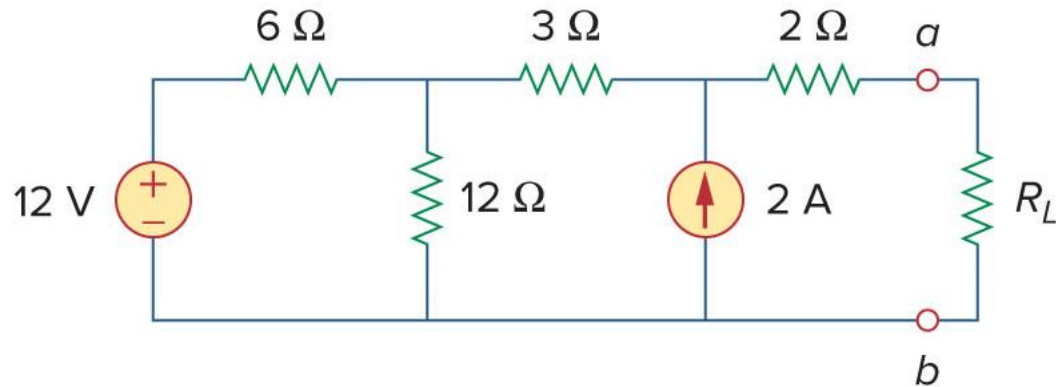
It can be shown that this point is the global **maximum extremum** since

$$\frac{d^2p}{dR_L^2} < 0$$

Exercise 10

Find the value of load resistance R_L for maximum power transfer in the circuit below, and then find the maximum power p .

Solution in Exercises_solution_Topic_3.pdf



Questions?

